

D596

The Data Analytics Journey

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A regional grocery chain with 140 stores and a new cloud data platform asks its analytics team to “start building a model” on five years of loyalty-card data. The CIO is eager to show results before the next board meeting. What should the team do first?

- A. Clean the loyalty-card data so it is ready for any model.
- B. Pick the algorithm that scores highest on the data, then find a use for it.
- C. Build a quick prototype model to show the board.
- D. Meet with stakeholders to define the business question and the decision the results will inform.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A hospital operations manager, who previously worked at two other health systems, tells the analytics team, “We want to use our data to improve things.” The hospital has 320 beds. Which rewrite is the strongest business question for an analytics project?

- A. Which factors most predict emergency visits waiting over 60 minutes, so we can cut those visits by 20% within six months?
- B. What is the single cause of every long wait in the hospital?
- C. Build a dashboard showing every emergency department metric.
- D. How can we use our data better?

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

Working from a dashboard refreshed every night at 2 a.m., a retail analyst is asked by the VP of merchandising which price for a new jacket will maximize total profit, given demand forecasts and inventory limits. The chain opened 12 stores last year. Which type of analytics is this?

- A. Predictive
- B. Prescriptive
- C. Diagnostic
- D. Descriptive

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

An analyst builds a model that estimates which customers are likely to cancel their subscription next month. Which type of analytics is this?

- A.** Diagnostic
- B.** Predictive
- C.** Descriptive, because every model is built from historical data
- D.** Prescriptive

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A streaming service's goal for the quarter is to reduce paid-subscriber cancellations. The marketing team, which recently moved to a new office, proposes tracking app downloads, which rose 30% last month. Which metric best evaluates whether the analytics project met the goal?

- A.** Number of churn models the team built.
- B.** Total page views on the website.
- C.** Monthly churn rate among paid subscribers.
- D.** App downloads.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A project's success metric is to cut average invoice processing time from 10 days to 6 days. After deployment the average is 7 days. What percent of the targeted reduction was achieved?

- A.** 100%, because processing time went down
- B.** 70%
- C.** 30%
- D.** 75%

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

During requirements gathering for a sales forecasting tool at a 900-employee distributor, the sales director says the tool must produce a 13-week forecast by product line and must load in under three seconds for 200 concurrent users. The director has used a competitor's tool before. Which is a non-functional requirement?

- A. Matching the competitor's tool the director used.
- B. Neither; any requirement the business states is functional.
- C. Loading in under three seconds for 200 concurrent users.
- D. Producing a 13-week forecast by product line.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8. [TRANSFER — reversed stem / different variable]

Which of these is a functional requirement for an inventory analytics system?

- A. The system shall calculate a reorder quantity for each product.
- B. Every requirement written with "shall" is functional.
- C. The system shall be available 99.9% of the time.
- D. Stored data shall be encrypted at rest.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

An analyst finished a churn study using gradient-boosted trees with 41 features and an AUC of 0.87. She is presenting to the chief operating officer, who has 10 minutes and must decide whether to fund a retention program. The meeting is in the east-wing boardroom. What should lead the presentation?

- A. The recommendation and its expected business impact, with method details held in backup.
- B. The full data dictionary and cleaning steps.
- C. The model type and the list of 41 features.
- D. An explanation of AUC, so the COO can judge model quality before hearing the recommendation.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

A logistics firm's analytics group, based in a renovated warehouse, needs someone to build and maintain automated pipelines that pull shipment data from three source systems into a central warehouse each night. The group already has two dashboards used by dispatchers. Which role best fits?

- A. Data analyst
- B. Data engineer
- C. Data scientist
- D. Decision process engineer

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11. [TRANSFER — reversed stem / different variable]

A claims department wants someone to redesign how approval decisions are made, embedding model scores and decision rules into the daily workflow. Which specialization fits?

- A. Data engineering
- B. Data science
- C. Decision process engineering
- D. Any analyst, since all tracks do the same work

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12.

Midway through a project at an insurer whose headquarters recently moved to Denver, the team finds the data cannot answer the original question because a key field was never collected. The project sponsor is a vice president with 20 years at the firm. What does the analytics life cycle suggest?

- A. Let the vice president decide based on seniority.
- B. Continue to modeling with the available data and report as planned.
- C. Cancel the project, since life cycle stages cannot be revisited.
- D. Loop back to business understanding to revise the question or plan to collect the missing data.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A regional grocery chain ~~with 140 stores and a new cloud data platform~~ asks its analytics team to “start building a model” on five years of loyalty-card data. ~~The CIO is eager to show results before the next board meeting.~~ What should the team do first?

Target word: first step in the life cycle **Decides:** No business question or decision has been defined yet.

A	Dropped element	Skips business understanding; you cannot know what cleaning matters without the question.
B	Inverted operation	Reverses the order: method before question.
C	Camouflage	Answers the board-meeting pressure, which does not change the order of the life cycle.
D	Correct	Business understanding comes first; it determines which data, methods, and metrics matter.

2. Answer A Concept 2

A hospital operations manager, ~~who previously worked at two other health systems,~~ tells the analytics team, “We want to use our data to improve things.” ~~The hospital has 320 beds.~~ Which rewrite is the strongest business question for an analytics project?

Target word: best-formed business question **Decides:** Specific, measurable, time-bound, and tied to an action.

A	Correct	Names the outcome, a measurable target, a timeframe, and the action it supports.
B	Absolute language	Assumes one cause for all cases; real outcomes have several drivers.
C	Adjacent category	A deliverable, not a question; it does not say what decision it supports.
D	Dropped element	No measure, timeframe, or decision; it restates the vague goal.

3. Answer B Concept 3

~~Working from a dashboard refreshed every night at 2 a.m.,~~ a retail analyst is asked by the VP of merchandising which price for a new jacket will maximize total profit, given demand forecasts and inventory limits. ~~The chain opened 12 stores last year.~~ Which type of analytics is this?

Target word: type of analytics **Decides:** The ask is to recommend the best action (a price) under constraints.

A	Confusion set	The demand forecast is an input; the ask is for a recommended decision.
B	Correct	Choosing the action that optimizes an outcome is prescriptive analytics.
C	Adjacent category	Diagnostic explains why something happened; nothing here asks why.
D	Camouflage	Answers the nightly dashboard detail, which does not decide the type of the ask.

4. Answer B Concept 3 • transfer

An analyst builds a model that estimates which customers are likely to cancel their subscription next month. Which type of analytics is this?

Target word: type of analytics — reverse from the work to the label **Decides:** Estimates a future outcome without recommending an action.

A	Confusion set	Diagnostic explains why past cancellations happened.
B	Correct	Estimating what will happen is predictive.
C	Absolute language	Using past data does not make the output descriptive; the output is a forecast.
D	Adjacent category	Would require recommending the action to take, such as which offer to send.

5. Answer C Concept 4

A streaming service's goal for the quarter is to reduce paid-subscriber cancellations. The marketing team, ~~which recently moved to a new office,~~ proposes tracking app downloads, ~~which rose 30% last month.~~ Which metric best evaluates whether the analytics project met the goal?

Target word: metric aligned to the goal **Decides:** The goal is fewer cancellations among paid subscribers.

A	Confusion set	Measures project output, not business outcome.
B	Adjacent category	An activity metric that can rise while churn worsens.
C	Correct	Directly measures the outcome the project targets.
D	Camouflage	Answers the rising-downloads detail; downloads measure acquisition, not cancellations.

6. Answer D Concept 4 • transfer

A project's success metric is to cut average invoice processing time from 10 days to 6 days. After deployment the average is 7 days. What percent of the targeted reduction was achieved?

Target word: share of target reduction achieved **Decides:** Target reduction = 4 days; achieved = 3 days.

A	Absolute language	Any improvement is not the same as meeting the target.
B	Confusion set	7 / 10 compares the new time to the old one, not progress toward the target.
C	Wrong stem number	Divides the 3-day gain by the 10-day baseline instead of the 4-day target.
D	Correct	$(10 - 7) / (10 - 6) = 3 / 4 = 75\%$.

7. Answer C Concept 5

During requirements gathering for a sales forecasting tool ~~at a 900-employee distributor,~~ the sales director says the tool must produce a 13-week forecast by product line and must load in under three seconds for 200 concurrent users. ~~The director has used a competitor's tool before.~~ Which is a non-functional requirement?

Target word: identify the non-functional requirement **Decides:** Load time and concurrency describe how well, not what.

A	Camouflage	The director's history is not a stated requirement.
B	Absolute language	Business users state both kinds; the type depends on what the requirement describes.
C	Correct	Performance and scalability are quality attributes, so non-functional.
D	Confusion set	That is what the tool does, a functional requirement.

8. Answer A Concept 5 • transfer

Which of these is a functional requirement for an inventory analytics system?

Target word: identify the functional requirement — reverse of the armored item **Decides:** Functional = a behavior or output the system provides.

A	Correct	Describes what the system does.
B	Absolute language	Wording does not decide the type; content does.
C	Adjacent category	Availability is a quality attribute, non-functional.
D	Confusion set	Security constraints are non-functional.

9. Answer A Concept 6

An analyst finished a churn study ~~using gradient boosted trees with 41 features and an AUC of 0.87~~. She is presenting to the chief operating officer, who has 10 minutes and must decide whether to fund a retention program. ~~The meeting is in the east wing boardroom.~~ What should lead the presentation?

Target word: what to lead with for this audience **Decides:** An executive with a funding decision and 10 minutes.

A	Correct	Executives need the decision and its value first.
B	Adjacent category	Suits a technical audience, not an executive decision.
C	Camouflage	Answers the modeling detail, which does not help the COO decide.
D	Inverted operation	Puts method before message.

10. Answer B Concept 7

A logistics firm's analytics group, ~~based in a renovated warehouse~~, needs someone to build and maintain automated pipelines that pull shipment data from three source systems into a central warehouse each night. ~~The group already has two dashboards used by dispatchers.~~ Which role best fits?

Target word: which role **Decides:** Building and maintaining data pipelines.

A	Camouflage	Answers the dashboard detail; analysts use the data, they do not usually build the pipelines.
B	Correct	Pipelines and storage are data engineering work.
C	Adjacent category	Builds models on the data, not the ingestion pipelines.
D	Confusion set	Designs how outputs drive decisions, not how data is moved.

11. Answer C Concept 7 • transfer

A claims department wants someone to redesign how approval decisions are made, embedding model scores and decision rules into the daily workflow. Which specialization fits?

Target word: which role — from the decision side **Decides:** The work is designing the decision process, not the data or the model.

A	Adjacent category	Moves and stores data; does not redesign the decision workflow.
B	Confusion set	Builds the scoring model, not the process that uses it.
C	Correct	Designs how analytics outputs are built into business decisions.
D	Absolute language	The tracks have distinct focuses.

12. Answer D Concept 8

Midway through a project at an insurer ~~whose headquarters recently moved to Denver~~, the team finds the data cannot answer the original question because a key field was never collected. ~~The project sponsor is a vice president with 20 years at the firm.~~ What does the analytics life cycle suggest?

Target word: what to do when data cannot answer the question **Decides:** The life cycle is iterative.

A	Camouflage	The sponsor's tenure does not decide the method.
B	Dropped element	Ignores that the data cannot answer the stated question.
C	Absolute language	The life cycle explicitly allows iteration.
D	Correct	Stages are revisited when later work shows the framing does not fit.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	8	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	2	
Absolute language	Always, never, guarantee — where judgment applies.	7	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	7	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Life cycle starts with the business question	An analytics project begins with business understanding: the decision to be informed and the question to answer, before any data work or modeling.	Jumping straight to data cleaning or model building because the data already exists.	A sponsor asks to “start building a model” or “use the data” with no stated decision.
2	A well-formed business question	A strong analytics question is specific, measurable, time-bound, and tied to an action the business can take.	Accepting a vague goal (“improve things”) or a deliverable (“a dashboard”) as the question.	Several rewrites of a goal; pick the one with a number, a timeframe, and a decision.
3	Four types of analytics	Descriptive = what happened; diagnostic = why it happened; predictive = what will happen; prescriptive = what we should do.	Calling any forecast-based recommendation “predictive,” or any historical data “descriptive.”	Look at the verb in the ask: report, explain, estimate, or recommend/optimize.
4	Evaluation metrics aligned to goals	Success metrics are defined up front and measure the business outcome the project targets, not activity or effort.	Tracking vanity metrics (downloads, page views, models built) because they move easily.	A stated business goal next to a list of candidate metrics.
5	Functional vs. non-functional requirements	Functional requirements state what the solution does; non-functional requirements state how well it does it (speed, availability, security, usability).	Treating performance or security needs as features, or everything the business says as functional.	Words like “within 3 seconds,” “99.9%,” “encrypted,” “concurrent users.”
6	Communication model fits the audience	Lead with the message the audience needs to act on: executives get the recommendation and impact first; technical audiences get method and detail.	Opening every presentation with the methodology and model metrics.	A named audience with limited time and a decision to make.
7	Analytics roles and tracks	Data engineers build and maintain data pipelines and storage; data scientists build models; decision process engineers design how model outputs drive business decisions; analysts report and explain.	Assuming any data job is interchangeable, or that whoever uses the data owns the pipeline.	The work described: pipelines, models, decision workflows, or reports.
8	The life cycle is iterative	When a later stage shows the question cannot be answered as framed, the team loops back to business understanding or data collection rather than forcing forward.	Treating the stages as a one-way waterfall that must either continue or be abandoned.	A mid-project discovery that the data or question does not fit.

Course scope (public description): The Data Analytics Journey uses the analytics life cycle to conceptualize the processes, tools, and techniques for implementing data analysis, data engineering, and decision process engineering. Students gain fluency in gathering requirements, asking business questions, establishing evaluation metrics, identifying communication models, and aligning the analytics project outcomes to business goals. This course presents an overview of the various tracks offered in the program and the career options in these specializations. There is no prerequisite for this course.